Production and Quality Control
Datasheet TeraFlash pro

1.0 General

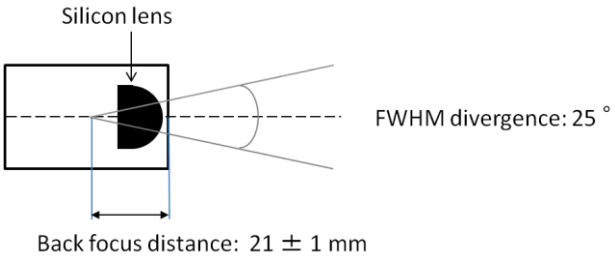
Order Number	VB.26.02551
Customer	Prof. Mi-a Seo
Institute / Company	Sogang University (via Jinsung Instruments, Inc.)
Production Date	07.09.2026

2.0 Components

TeraFlash	TeraFlash pro HP 15-110 cm usable THz path length	SN TF7-2303
Laser	FF smart THz	SN Beta3
Tx Antenna Modules	PCA-TD-250-TX-4 High-Power InGaAs Photoconductive Switch	SN FTP250328B
Rx Antenna Modules	PCA-TD-250-RX-4 InGaAs Photoconductive Switch	SN RP230538

Antenna Package
Si lens, cylindrical housing, Ø 25 mm
Electrical connector: TX: Lemo, RX: SMB
Fiber: FC/APC connector, SM/PM fiber, 2.5 m length

Beam Properties
Linear THz polarization



Operating Conditions
Excitation wavelength: 1.5 µm
Mean optical power: 50 mW
Absolute maximum optical power: 60 mW
Bias voltage TX: 0 .. -200 V (unipolar)
Bias voltage RX: -3 .. +3 V (only for testing)

CAUTION!	**Photoconductive antennas are electrostatic-discharge sensitive devices!** **We recommend to always wear a high-impedance grounding strap for handling.**
CAUTION!	**Clean fiber facets before connecting the fibers.**

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3.0 Acceptance Criteria

	Specification	Production check Channel 1	Production check Channel 2
Peak Dynamic Range (2 averages)	typ. 70 dB	90 dB	89 dB
Bandwidth (2 averages)	typ. 4.5 THz	4.5 THz	4.5 THz
Peak Dynamic Range (1000 averages)	> 100 dB	116 dB	116 dB
Bandwidth (1000 averages)	6 THz	6.2 THz	6.2 THz
Peak Dynamic Range (10000 averages)	> 110 dB	125 dB	125 dB
Bandwidth (10000 averages)	6 THz	6.3 THz	6.3 THz

4.0 Measurements

Measurement Details TIA-Conditions: ± 4000 nA

Values Measured **Optical power**:

TX1 Output: 36 mW after output port

RX1 Output: 13 mW after output port

TX2 Output: 36 mW after output port

RX2 Output: 13 mW after output port

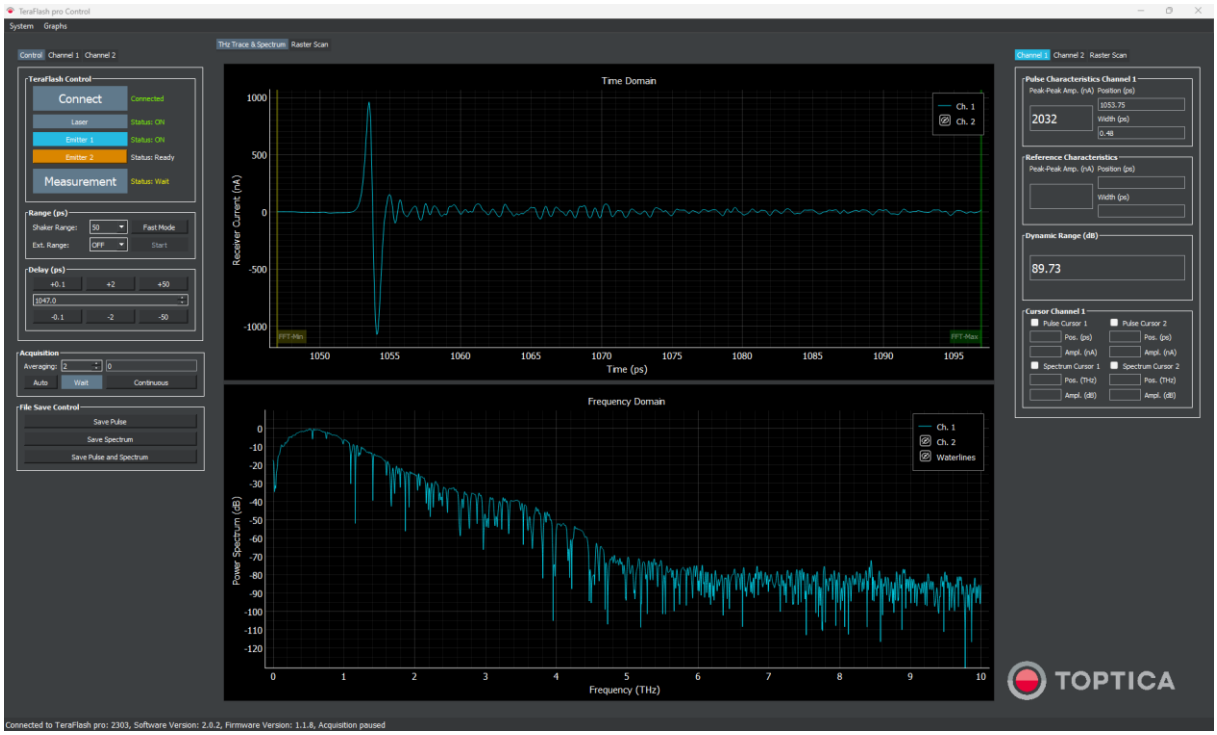
Terahertz Reflection Spectrum, Channel 1: Optomechanical setup: Imaging Extension in reflection

4 mirrors, 2” focal length (coll.), 4” focal length (focusing), 1” diameter

Terahertz pulse position: 1055 ps (free-space path approx. 32 cm)

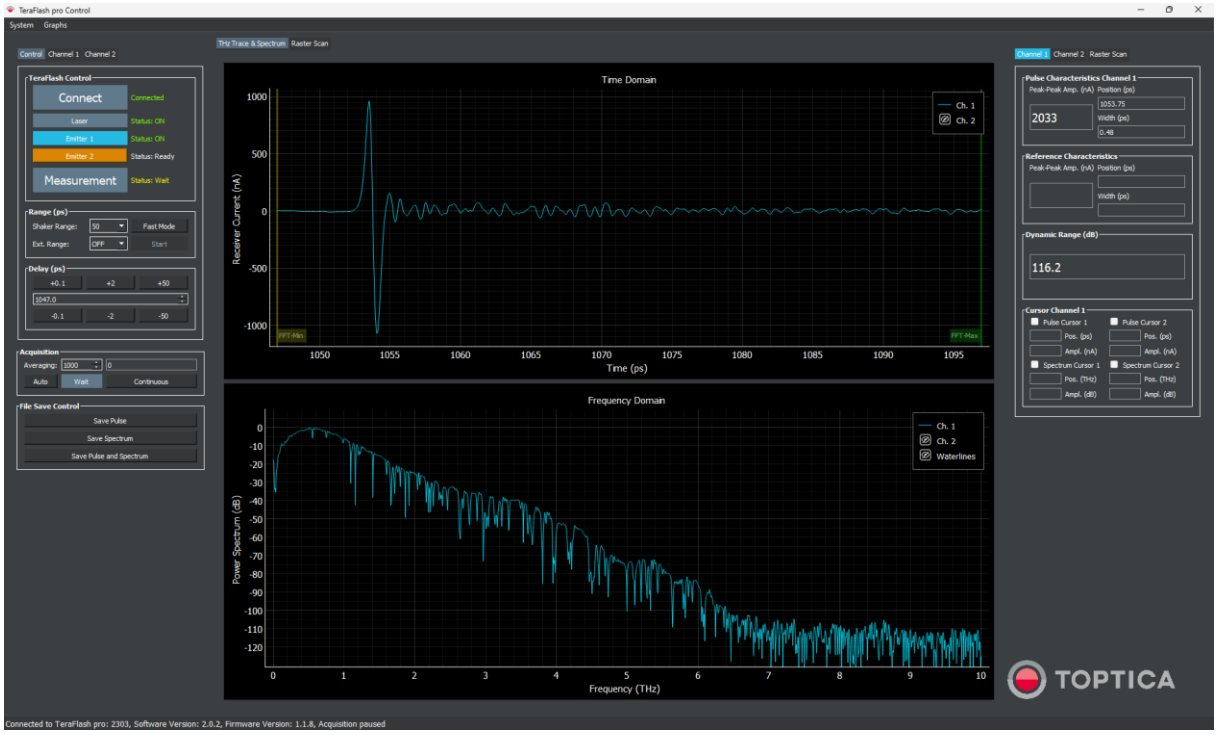
Peak-Peak-Amplitude: 2040 nA

2 traces averaged

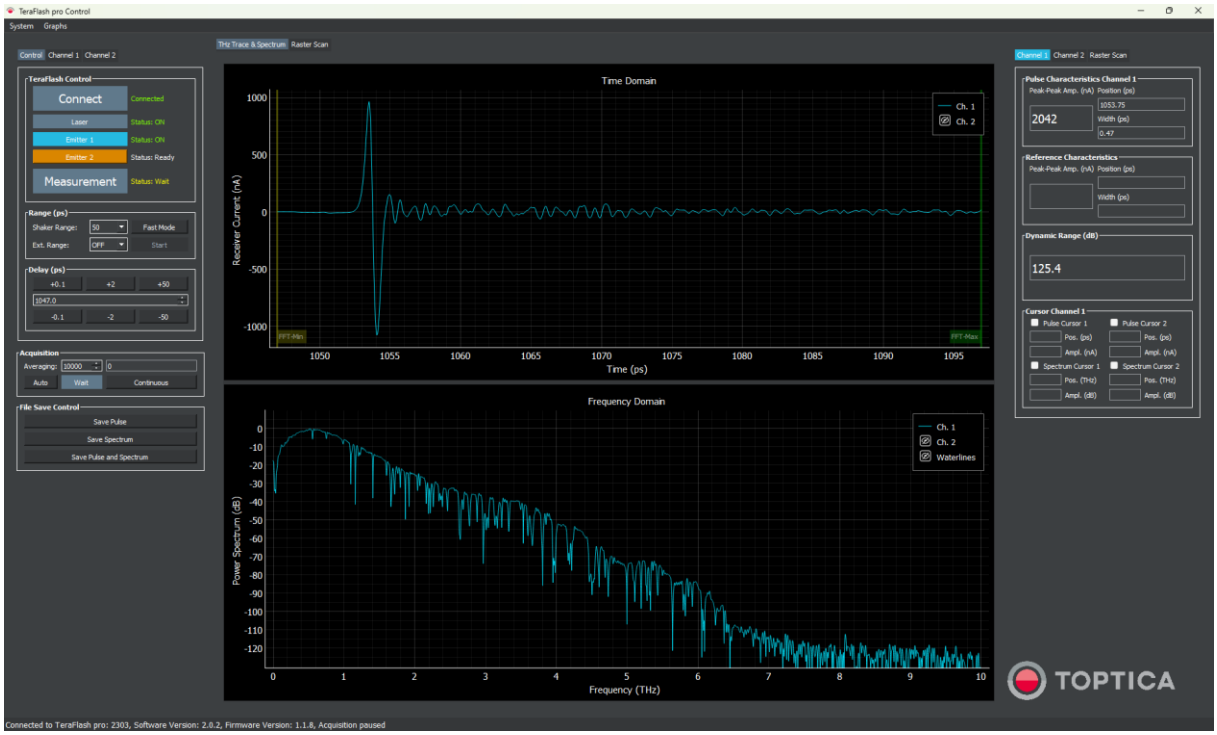


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1000 traces averaged



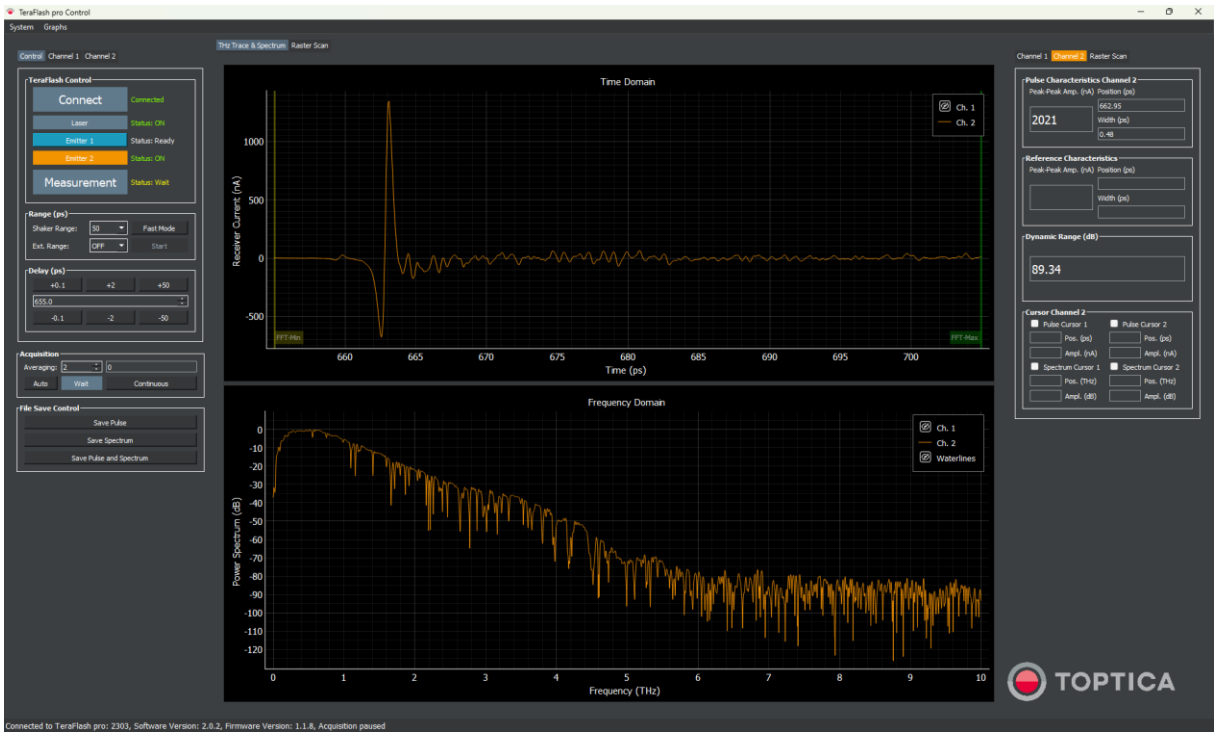
10000 traces averaged



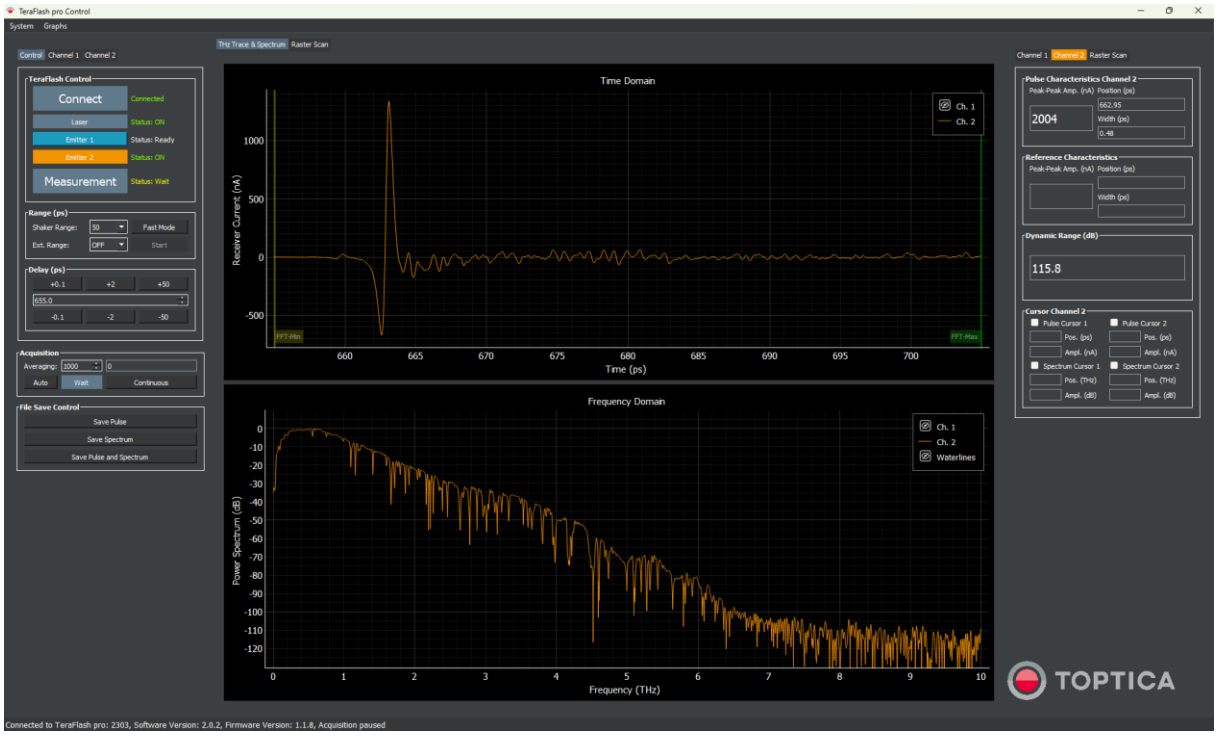
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Terahertz Transmission Spectrum, Channel 2: Optomechanical setup: reflection optics THz Optics / Tr 2M
2 mirrors, 3” focal length (coll.), 2” diameter
Terahertz pulse position: 663 ps (free-space path approx. 24 cm)
Peak-Peak-Amplitude: 2020 nA

2 traces averaged

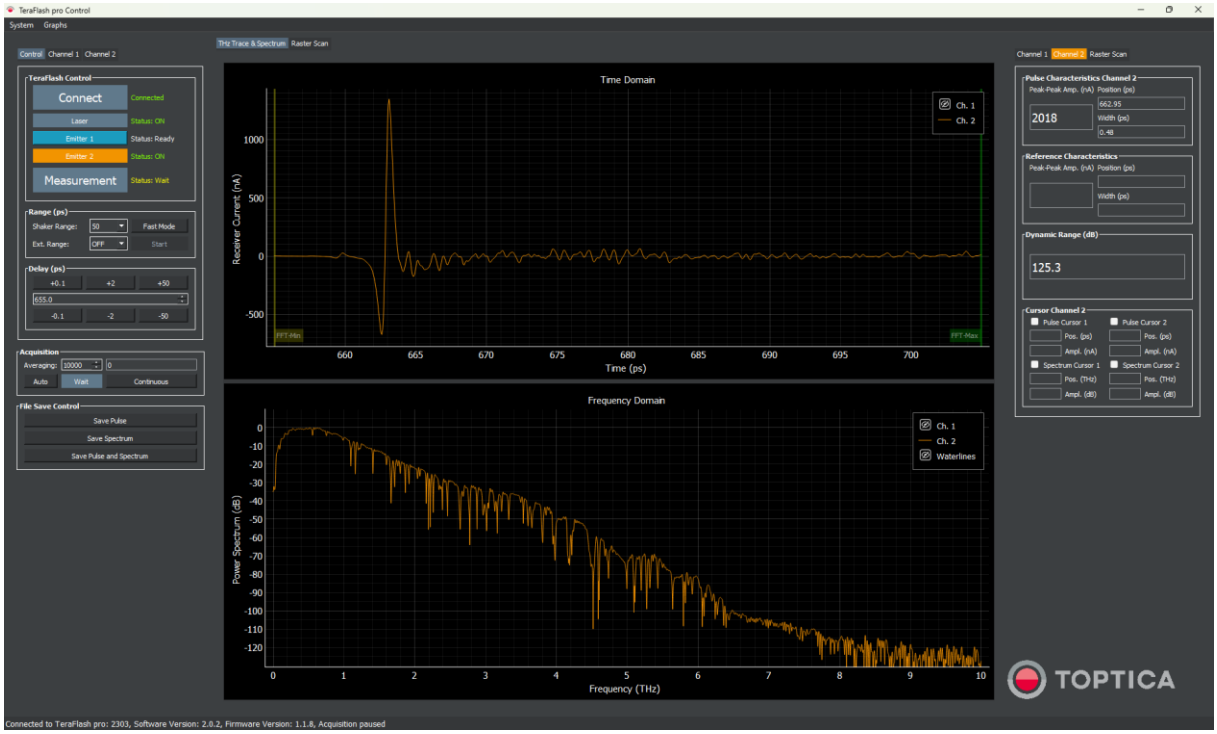


1000 traces averaged



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10000 traces averaged



5.0 Quality Control

Production	**Thomas Huber**	**Date**	**07.09.2026**
Final Check	**Stephan Reschke**	**Date**	**07.09.2026**

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